

Heart_Failure_Prediction (1).ipynb

colab.research.google.com/drive/1HUwSKMYev94e8mjD4ZFOsLHbGUWe9tM-?hl=en#scrollTo=QCX0Srm8WTzO

Heart_Failure_Prediction (1).ipynb

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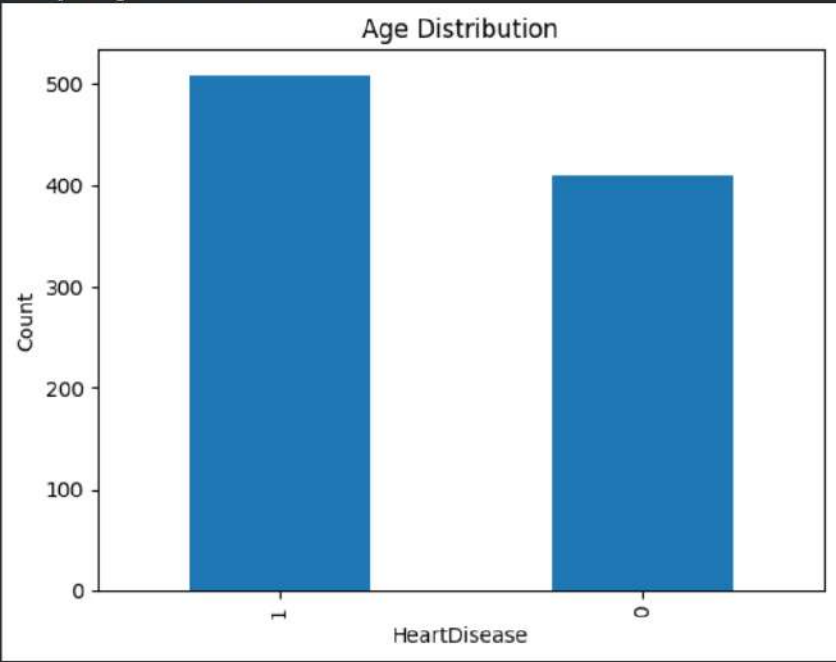
Commands + Code + Text Run all

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dtypes: float64(1), int64(6), object(5)
memory usage: 86.2+ KB

Age Distribution



HeartDisease	Count
1	510
0	410

Data Visualization

Variables Terminal

Heart_Failure_Prediction (1).ipynb

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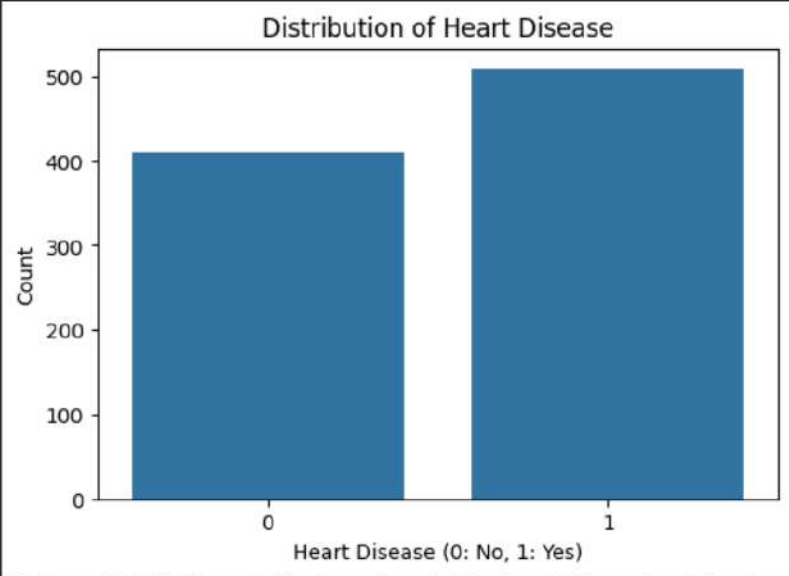
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```
[ ] ▶ sns.countplot(x='HeartDisease', data=df)
plt.title('Distribution of Heart Disease')
plt.xlabel('Heart Disease (0: No, 1: Yes)')
plt.ylabel('count')
plt.show()
print("Plot 1: Distribution of the target variable 'HeartDisease'. This shows the balance of classes.")
```

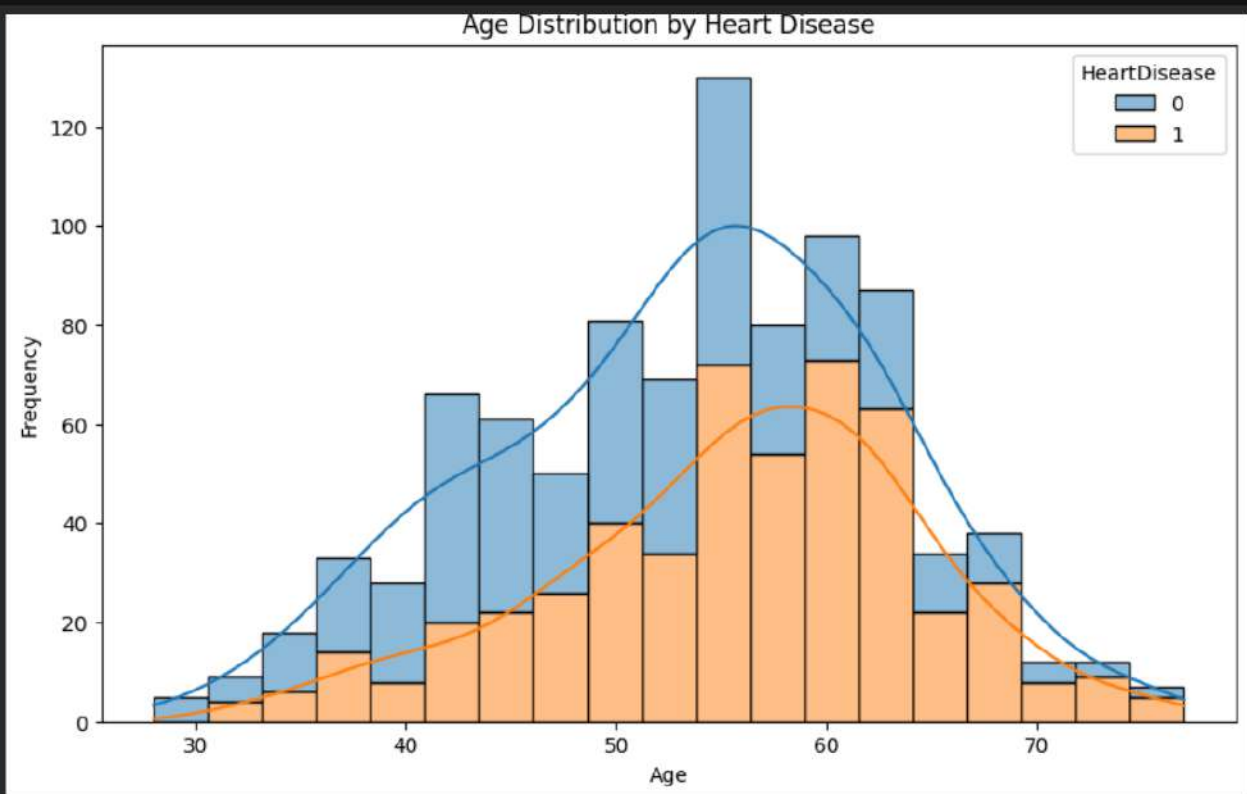
...



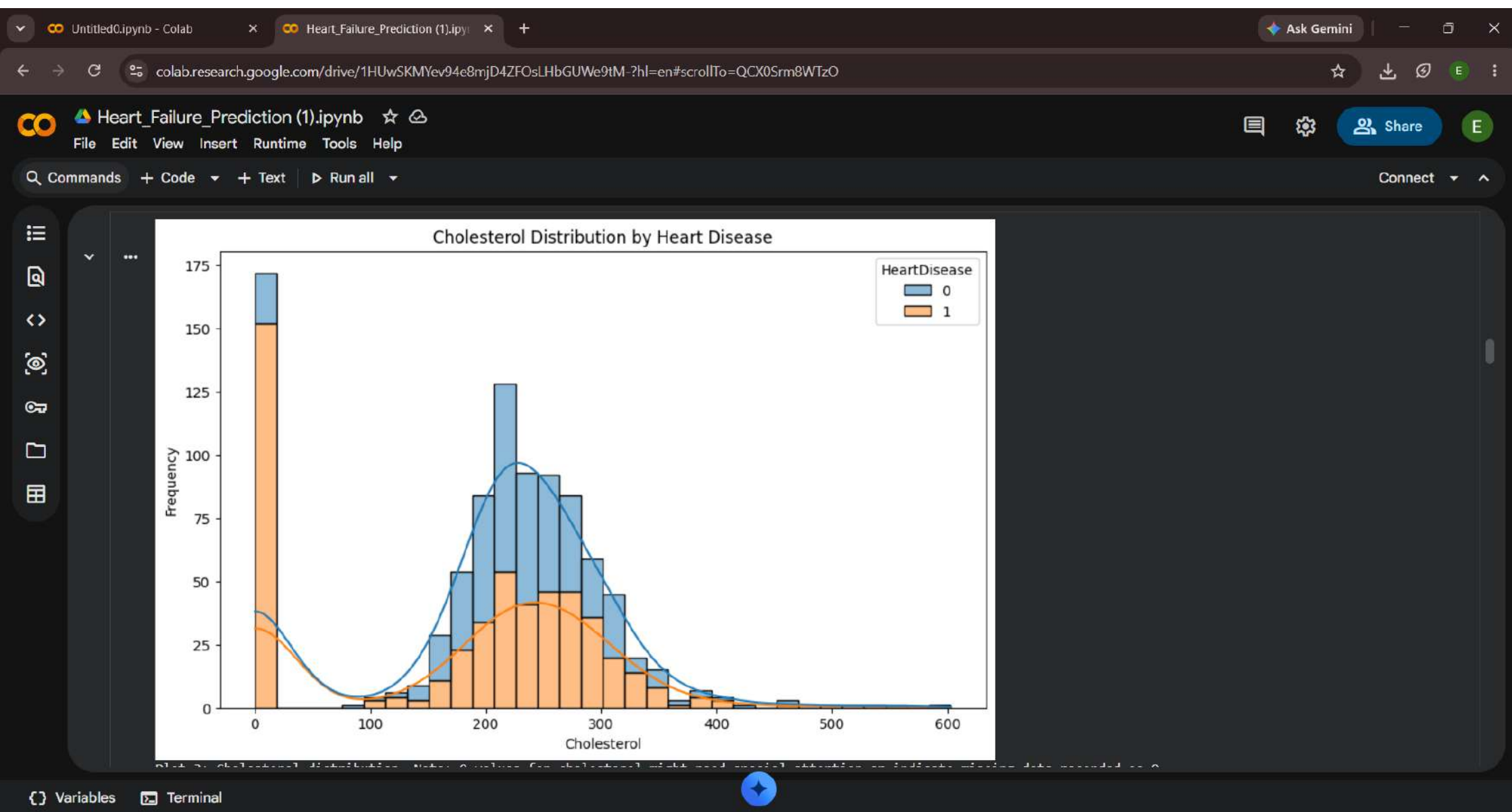
Heart Disease	Count
0	410
1	510

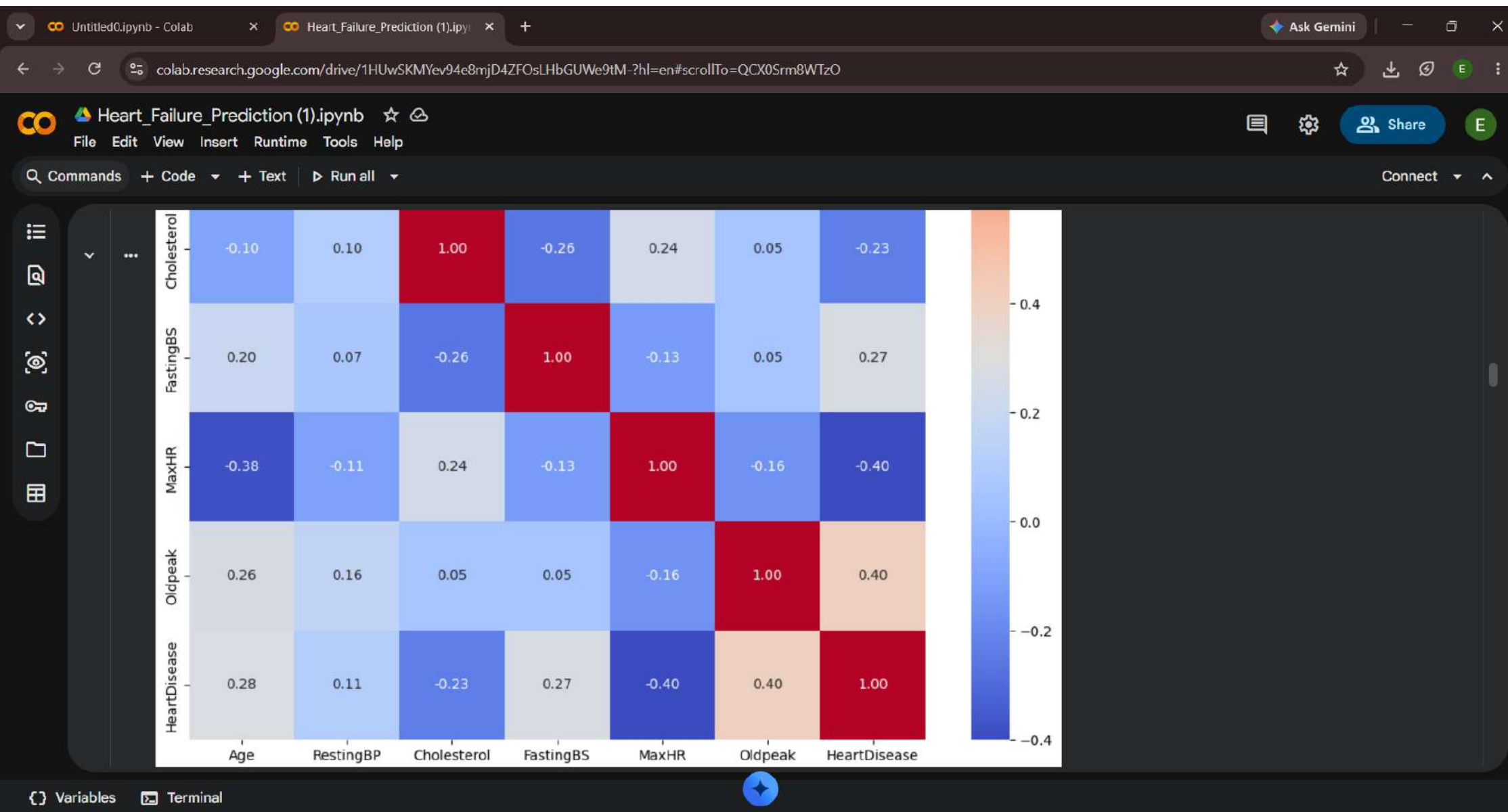
Plot 1: Distribution of the target variable 'HeartDisease'. This shows the balance of classes.

Variables Terminal



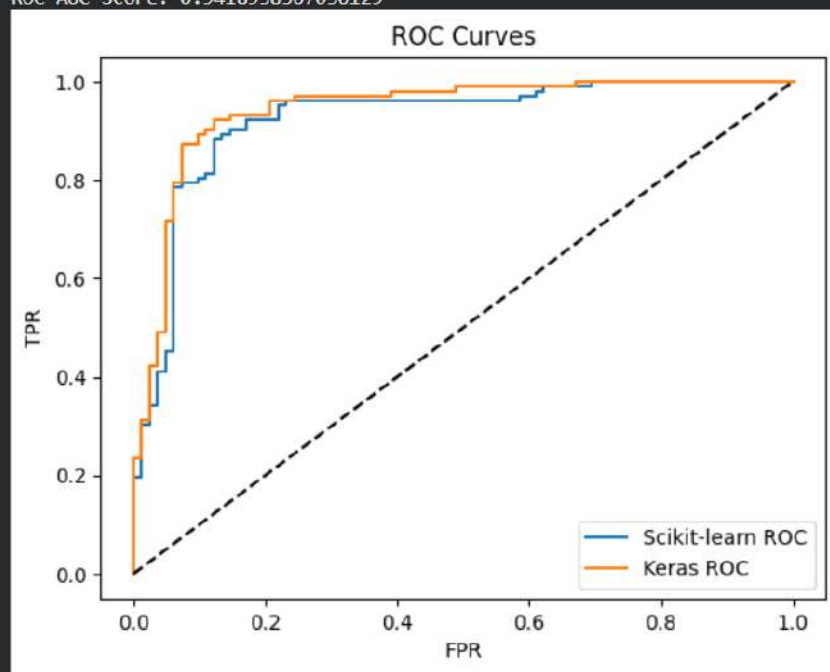
Plot 2: Age distribution for patients with and without heart disease.



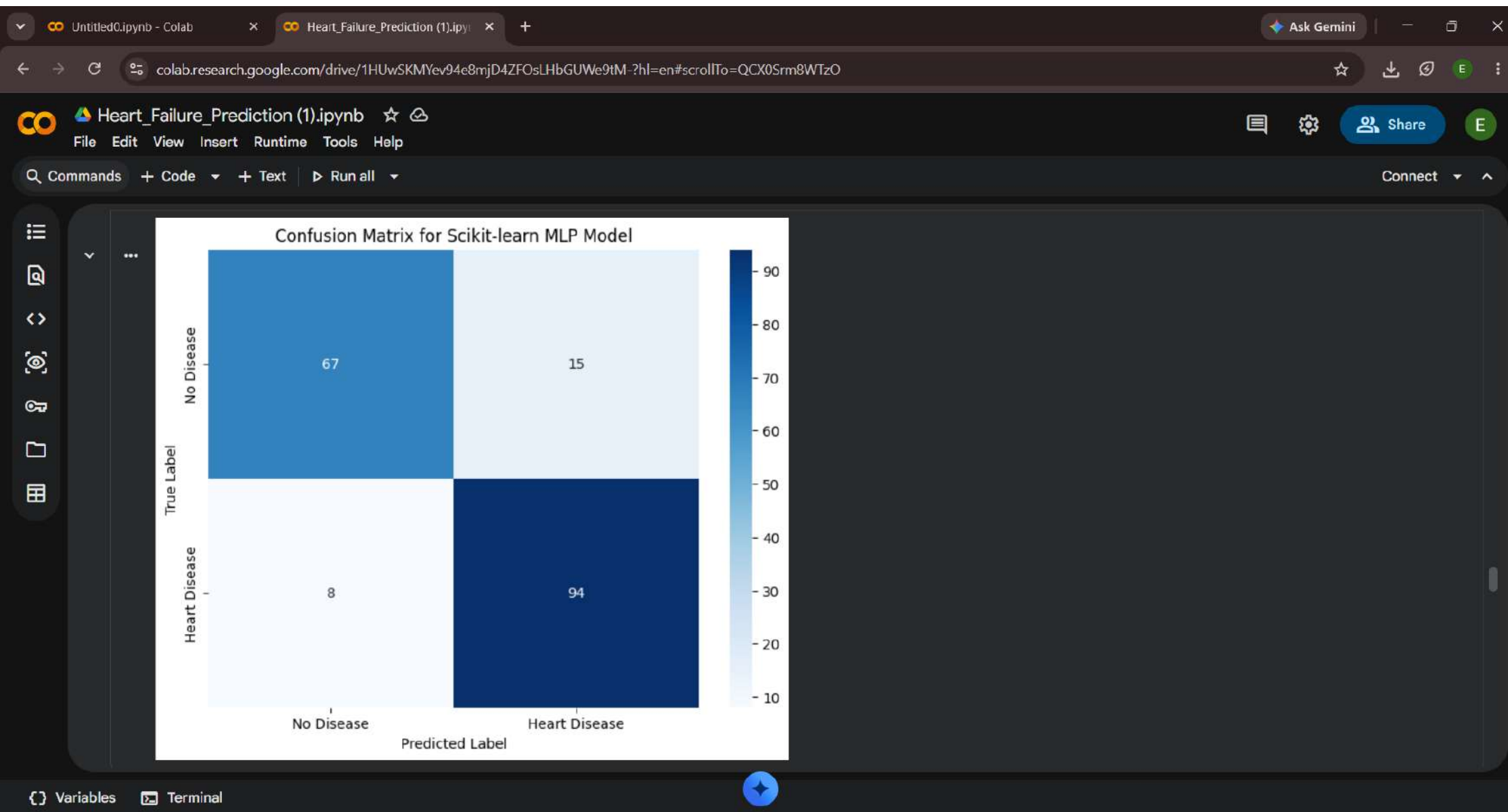


weighted avg 0.90 0.90 0.90 184

ROC AUC Score: 0.9418938307030129



```
[ ] # --- 3. Experimenting with Scikit-learn MLP parameters ---  
print("\n 3. Experimenting with Scikit-learn MLP parameters")
```



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Ask Gemini

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print("\nKeras Model (Tuned) - Summary:")

keras_model_tuned.summary()

...

Keras Model (Tuned) - Summary:

Model: "sequential_1"

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 128)	2,688
dropout (Dropout)	(None, 128)	0
dense_4 (Dense)	(None, 64)	8,256
dropout_1 (Dropout)	(None, 64)	0
dense_5 (Dense)	(None, 32)	2,080
dense_6 (Dense)	(None, 1)	33

Total params: 13,057 (51.00 KB)
Trainable params: 13,057 (51.00 KB)
Non-trainable params: 0 (0.00 B)

[]

Callbacks for more advanced training

reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=5, min_lr=0.00001, verbose=0)

The 'patience' and 'factor' for ReduceLROnPlateau are among the parameters to tune.

factor: factor by which the learning rate will be reduced. new_lr = lr * factor.

patience: number of epochs with no improvement after which learning rate will be reduced.

Variables Terminal

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Scikit-learn Model:

	precision	recall	f1-score	support
0	0.91	0.83	0.87	82
1	0.87	0.93	0.90	102
accuracy			0.89	184
macro avg	0.89	0.88	0.88	184
weighted avg	0.89	0.89	0.89	184

AUC: 0.93675274988044

6/6 0s 8ms/step

Keras Model:

	precision	recall	f1-score	support
0	0.90	0.87	0.88	82
1	0.90	0.92	0.91	102
accuracy			0.90	184
macro avg	0.90	0.89	0.90	184
weighted avg	0.90	0.90	0.90	184

AUC: 0.9418938307030129

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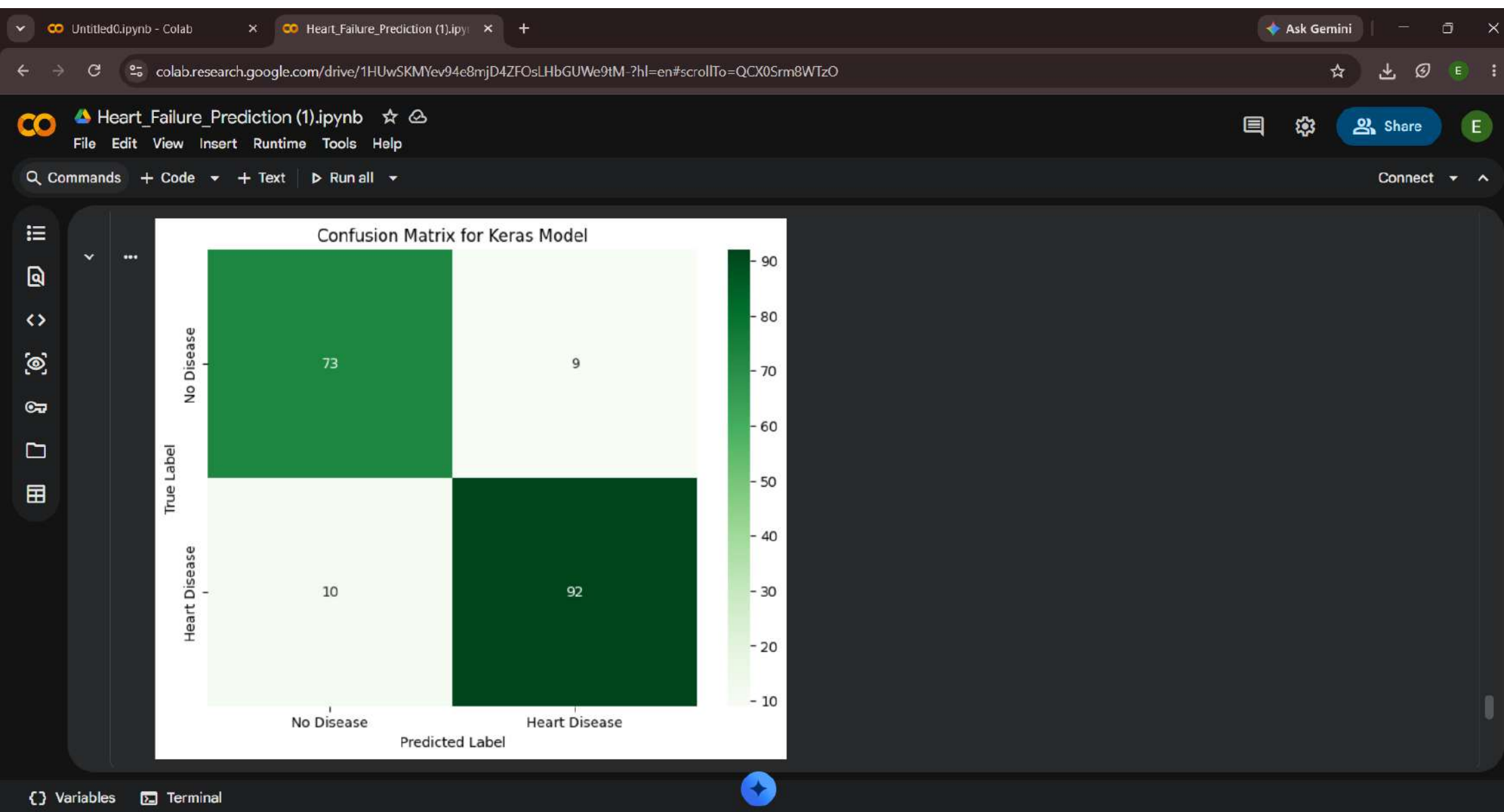
Convert probabilities to binary predictions

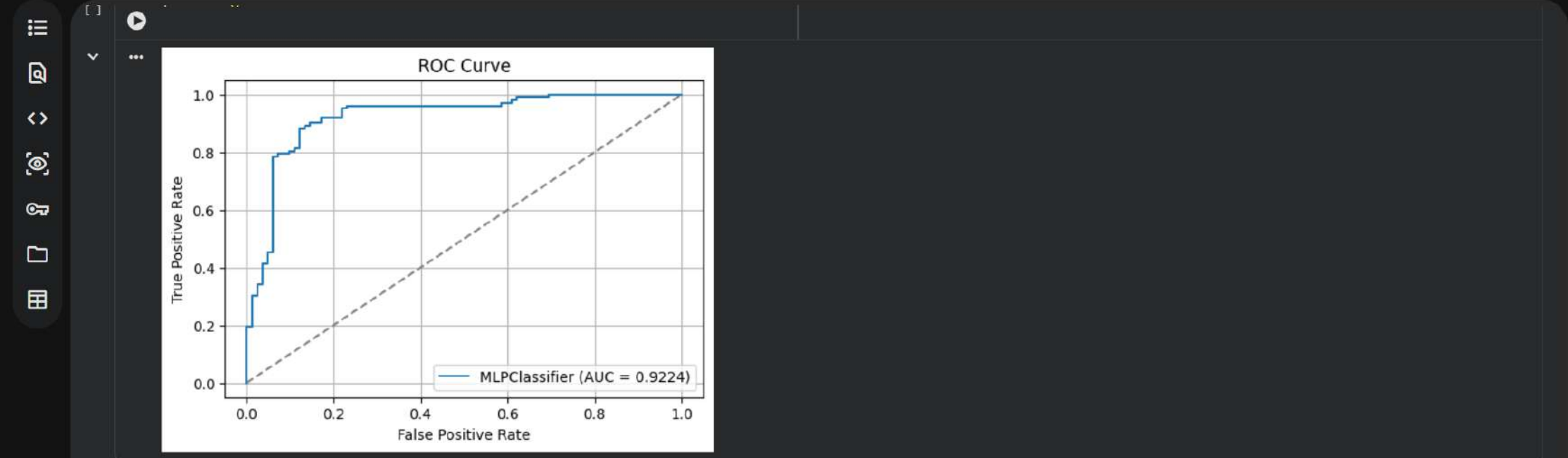
print("Keras Model (Tuned) - Classification Report:")

print(classification_report(y_test, y_pred_keras_tuned))

print("Keras Model (Tuned) - Confusion Matrix:")

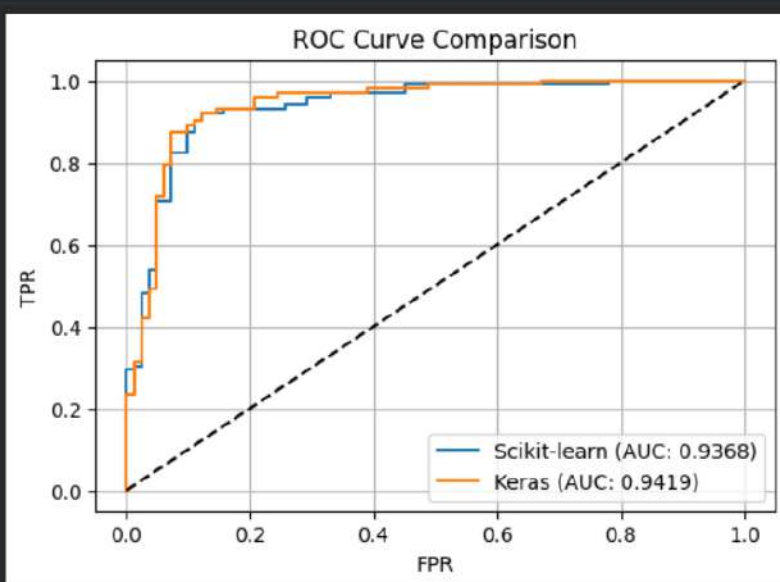
Variables Terminal





```
[ ]  
fpr1, tpr1, _ = roc_curve(y_test, y_prob_sk)  
fpr2, tpr2, _ = roc_curve(y_test, y_pred_keras)  
  
plt.figure(figsize=(6, 4))  
plt.plot(fpr1, tpr1, label='Scikit-learn (AUC: {:.4f})'.format(roc_auc_score(y_test, y_prob_sk)))
```

```
[ ] plt.show()
```



Final Observations:

- Both models performed well on the dataset.